

# Rohit Ganguly

Full-Stack & AI Systems Engineer — React/Next.js, NestJS, RAG & LLM Integration

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## Education

### Manipal University Jaipur

Aug 2023 – May 2027

B.Tech in Computer Science & Engineering

- CGPA: 9.06 (Dean's List) | Coursework: Data Structures & Algorithms, Operating Systems, DBMS, Computer Networks, AI

## Work Experience

### Software Engineering Intern — Betatest Solutions Pvt. Ltd.

Nov 2025 – Apr 2026

React Native (Expo) · Next.js · NestJS · TypeScript · PostgreSQL · Redis · Azure DevOps · k6 · New Relic

- Shipped **4 production features** (Settings, Edit Profile, biometric authentication, image picker) in React Native/Expo to iOS and Android, used daily by live end users.
- Designed and built **modular REST APIs in NestJS** for activity-type filtering, CSV report export, automated soft-delete cleanup, and cron-based task scheduling.
- Engineered a full **password reset and OTP verification flow** end-to-end (token validation, transactional email dispatch, Next.js UI) spanning frontend and backend.
- Implemented **i18n support (English, PT-BR)** and push/email notification pipelines, extending the product to multi-region users.
- Improved backend reliability via **database schema migrations and k6 load testing**; monitored production health with New Relic and Azure Pipelines CI/CD; **merged 55+ PRs** across a mono-repo.

### Full-Stack Developer Intern — Coding Jr

Jun 2025 – Aug 2025

VS Code Extension API · Next.js · Node.js · Redis · Figma

- Built **Planto, a VS Code and Eclipse extension**, end-to-end — chat interface, agent mode, image-to-code, and Figma-to-code generation workflows.
- Engineered **@file workspace attachment with Redis caching** for low-latency context handling in the AI coding assistant, and redesigned the Planto marketing site with Figma-driven UI/UX improvements.
- Contributed to an **AI-assisted resume builder** with section duplication, inline bullet editing, and LLM-powered formatting suggestions.

## Projects

### DevPulse — AI-Powered Developer Productivity Dashboard

github.com/psychic-coder/DevPulse

Next.js · NestJS · TypeScript · PostgreSQL · Redis · Socket.IO · Python (Flask) · Docker · k6 · Anthropic API

- Built a **full-stack analytics platform** that ingests a developer's GitHub history via OAuth, with a **Redis-backed caching layer** that tracks rate-limit headers to stay under API quotas while serving sub-second dashboard loads.
- Implemented a **real-time activity feed with Socket.IO**, streaming live commit and PR events to the Next.js dashboard without polling or page refresh.
- Integrated the **Anthropic Claude API** to auto-generate weekly developer digests and score pull-request quality (1–10) from title/body analysis, returned as structured JSON.
- Built a **Python (Flask) microservice** with Pandas/NumPy for commit-streak and language analytics, plus **OS Finder** — a personalized repo discovery engine ranking projects via a custom **New Contributor Friendliness (NCF) score**; containerized with Docker, load-tested with k6.

### CodeSage — Graph-Augmented Repository Intelligence Platform

github.com/psychic-coder/CodeSage

Python · FastAPI · Neo4j · Qdrant · OpenAI API · LangChain · tree-sitter · Celery · Redis · Next.js · Docker

- Architected a **Graph-Augmented RAG pipeline** combining Qdrant vector search (1536-dim embeddings) with Neo4j BFS traversal (up to 3 hops) to predict file-level change impact across **10+ languages**, ranked by semantic similarity, graph centrality, and risk score.
- Built a **multi-stage async ingestion pipeline** using tree-sitter AST parsing to resolve imports, function calls, and class hierarchies into a typed Neo4j dependency graph, processing **1,000-file repositories in under 3 minutes** via batched Cypher UNWIND queries.
- Engineered a **weighted risk-scoring algorithm** (NetworkX centrality, cyclomatic complexity) to surface circular dependencies and god files; offloaded ingestion to **Celery + Redis pub/sub** for real-time progress streaming, reaching **87% impact-prediction accuracy**.

## Technical Skills

**Languages:** Java, JavaScript (ES6+), TypeScript, Python, C, SQL

**Frontend:** React.js, Next.js (TypeScript), React Native (Expo), TailwindCSS, Material-UI, Framer Motion

**Backend:** Node.js, Express.js, NestJS, REST APIs, Socket.IO, JWT Auth, Flask, FastAPI

**Databases & AI:** PostgreSQL, MongoDB, Redis, Neo4j, Qdrant · LangChain, Anthropic & OpenAI APIs, Pandas

**Tools & DevOps:** Git, Docker, Azure DevOps, GitHub Actions, k6, New Relic, Postman, Figma

## Certifications

Design and Analysis of Algorithms — **NPTEL** (Apr 2025) | Red Hat System Administration I & II (RH124, RH134) — **Red Hat** (Mar 2025) | Database Programming with SQL — **Oracle** (Nov 2024) | DSA using Java — **NPTEL** (Nov 2024)

## Honors & Awards

- 2nd Runner-Up** — BITS Pilani × Postman API Hackathon 4.0 (out of 1,000+ teams)
- Dean's List** (9.06 GPA) · **Student Excellence Award** (3×) — Manipal University Jaipur